Experiment #8
The Z-Transforms

1. Objective:

1.1 To study the Z-transform of aperiodic signals.

1.2 To Determine response of LTI system.

In this lab you learn

Poles, zeros and ROC of a LTI system calculation in python

2. Overview

In last lab, we discussed the discrete-time Fourier transform approach for representing discrete signals using complex exponential sequences. There are ***two*** shortcomings to the Fourier transform approach.

a) There are many useful signals in practice such as ***u[n]*** and ***nu[n]*** for which the discrete-time Fourier transform does not exist.

b) Second, the transient response of a system due to initial conditions or due to changing inputs cannot be computed using the discrete-time Fourier transform approach.

We now consider the z-transform an extension of the discrete-time Fourier transform to address the above two problems.

2.1 Definition

For a given sequence x[n], its z-transforms X[z] is defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \qquad (1)$$

Where z is a complex variable. If we let $z = re^{j\varpi}$, where' w' is a real frequency variable and r is attenuation then Equation (1) become

$$X(re^{j\varpi}) = \sum_{n=-\infty}^{\infty} x[n]r^{-n}e^{-nj\varpi} \qquad (2)$$

Which can be interpreted as the DTFT of a modified sequence {x[n]r-1}. For r=1(i.e. |z|=1), the z-transform of x[n] reduces to its DTFT. From equation (2), we see that $X(re^{j\varpi})$ is the Fourier transform of sequence of x[n] multiplied by real exponential r-n, i.e.

$$X(re^{j\varpi}) = \mathrm{F}\{x[n]r^{-n}\} \qquad (3)$$

2.2 Region of Convergence (ROC)

The range of z values for which X(z) exists is called the region of convergence (ROC) and is given by:

R-x < |z| < R+x (where, R+x and R-x positive numbers)

Convergence of series in Eq. (1) depends only on |z|, since |X(z)| < ∞ if,

$$\sum_{n=-\infty}^{\infty} |x[n]| \ |z^{-n}| \ < \ \infty \qquad (4)$$

The region of convergence consists of all values of z such that inequality (4) holds. Thus if some value of z, say z=z1 is in the ROC, then all values of z on the circle defined by |z|=|z1| will also be in the ROC. The ROC consists of a ring in the z-plane centered about the origin. Its outer boundary will be either a circle or may extend outward to infinity, and its inner boundary will be either a circle or may extend inward to include the origin. This is illustrated in Figure 1. In this figure X(z) will exist for all values of z lie in shaded region.

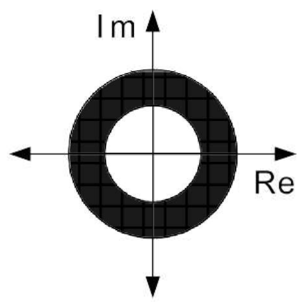

Figure 1

Following three points decide the shape of ROC

- The ROC for right-sided sequences (e.g. anu[n]) is always outside of a circle toward infinity.
- The ROC for left-sided sequences (e.g. anu[-n]) is always inside of a circle to include the origin.
- The ROC for two-sided sequences (e.g. anu[n]- anu[-n-1])) is always an open ring

2.3 **Z-Transforms of LTI system**

In case of LTI Discrete-time systems that we are concerned with in the text, all pertinent z-transforms are rational function of z-1.as expressed in flowing Eq. (5)

$$Y(z)=\frac{P(z)}{D(z)}=\frac{p_0+p_1z^{-1}+p_2z^{-2}+............+p_{M-1}z^{-(M-1)}+p_Mz^{-M}}{d_0+d_1z^{-1}+d_2z^{-2}+............+d_{N-1}z^{-(N-1)}+d_Nz^{-N}} \qquad (5)$$

p are numerator coefficients and q are denominator coefficients. Eq (5) results from solving LTI system in z domain.

3. Z-Transforms using Python:

3.1 [z,p,k]=tf2zp(num, den) command

Pole, zeros and ROC of the expression of following form can determine using tf2zp command. Consider response of an LTI system

$$G(z)=\frac{2z^4+16z^3+44z^2+56z+32}{3z^4+3z^3-15z^2+18z-12}$$

To find poles and zeros fist we carry out the factorization of G(z). Following code do this.

```
% Determination of factor form
num = input('Type the Numerators coefficients = ');
import numpy as np
from scipy import signal

# Define the numerator and denominator coefficients of the transfer function
num = [2 ,16, 44, 56, 32]
den = [3, 3, -15 ,18 ,-12]

# Calculate the z-transform
z, p, k = signal.tf2zpk(num, den)

# Calculate the ROC
roc = signal.convolve(den, den[::-1])

# Print the results
print("Transfer function coefficients:")
print("Numerator: ", num)
print("Denominator: ", den)
print("\nZeros: ", z)
print("Poles: ", p)
print("Gain: ", k)
print("\nROC: ", roc)
Output
zeros are at
  -4.0000
  -2.0000
  -1.0000 + 1.0000i
  -1.0000 - 1.0000i

poles are at
  -3.2361
```

1.2361
0.5000 + 0.8660i
0.5000 - 0.8660i

Gain Constant
0.6667

Radius of Poles
3.2361
1.2361
1.0000
1.0000

Zero-Pole Placement in Filter Design

LTI systems, particularly digital filters, are often designed by positioning zeros and poles in the z-plane. The strategy is to identify passband and stopband frequencies on the unit circle and then position zeros and poles by considering the following.

1. **Conjugate symmetry**

All poles and zeros must be paired with their complex conjugates.

2. **Causality**

To ensure that the system does not depend on future values, the number of zeros must be less than or equal to the number of poles.

3. **Origin**

Poles and zeros at the origin do not affect the magnitude response.

4. **Stability**

For a stable system, poles must be inside the unit circle. Pole radius is proportional to the gain and inversely proportional to the bandwidth. Passbands should contain poles near the unit circle for larger gains.

5. **Minimum phase**

Zeros can be placed anywhere in the z-plane. Zeros inside the unit circle ensure minimum

Phase. Zeros on the unit circle give a null response. Stopbands should contain zeros on or near the unit circle.

6. **Transition band**

A steep transition from passband to stopband can be achieved when stopband zeros are paired with poles along (or near) the same radial line and close to the unit circle.

7. **Zero-pole interaction**

Zeros and poles interact to produce a composite response that might not match design goals. Poles closer to the unit circle or farther from one another produce smaller interactions. Zeros and poles might have to be repositioned or added, leading to a higher filter order.

3.2 [r,p,k]=residuez(num,den) command

The M-file residuez can be used to develop partial fraction expansion of Eq (5) and vice versa.

$$G(z)=\frac{18z^3}{18z^3+3z^2-4z-12}$$

Now, we will find the partial fraction expansion using following code,

%Partial fraction expansion of rational z-transform

Output data

<u>Residues</u>

0.3600 0.2400 0.4000

<u>Poles</u> are at

0.5000 -0.3333 -0.3333

Note also that the z-transform of G(z) has double poles at z=-1/3. The first entry in both the residues and pole given above corresponds to the simple pole factor (1-0.5z-1), the second entry corresponds to the simple pole factor (1-0.333z-1), and the third entry correspond to the factor (1+0.333z-1) in the partial fraction. Thus the desired expansion is given by

$$G(z)=\frac{0.36}{1-0.5z^{-1}}+\frac{0.24}{1+0.33z^{-1}}++\frac{0.4}{(1+0.33z^{-1})^2}$$

Exercise:

P-1) Consider the following difference equations:

Obtain the pole zero plot of the given systems, Analyze the above systems for stability, find their Fourier transforms if possible. Compute the impulse response of the systems.

P-2) Find h[n]

1) $$H_2(z) = \frac{2z^4 + 6.4z^3 + 4.9z^2 - 0.1z - 0.6}{5z^4 + 15.5z^3 + 13.7z^2 - 22.52z - 0.48}$$

2) $$H_1(z) = \frac{(z+0.4)(z-0.91)(z^2+0.3z+0.4)}{(z^2-0.6z+0.6)(z^2+3z+5)}$$

Where powers of z are in negative, i.e. z^4 is actually z^{-4} etc. etc.